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EXPERIMENT 8

```
clc;
```

```
clear;
```

```
close all;
```

```
% Steering Angles (degrees)
```

```
angle = [-60 -30 0 30 60];
```

```
% Normalized Gain
```

```
gain = ones(size(angle));
```

```
% Create Polar Plot
```

```
figure;
```

```
polarplot(deg2rad(angle), gain, 'o-', 'LineWidth', 2);
```

```
% Add Title
```

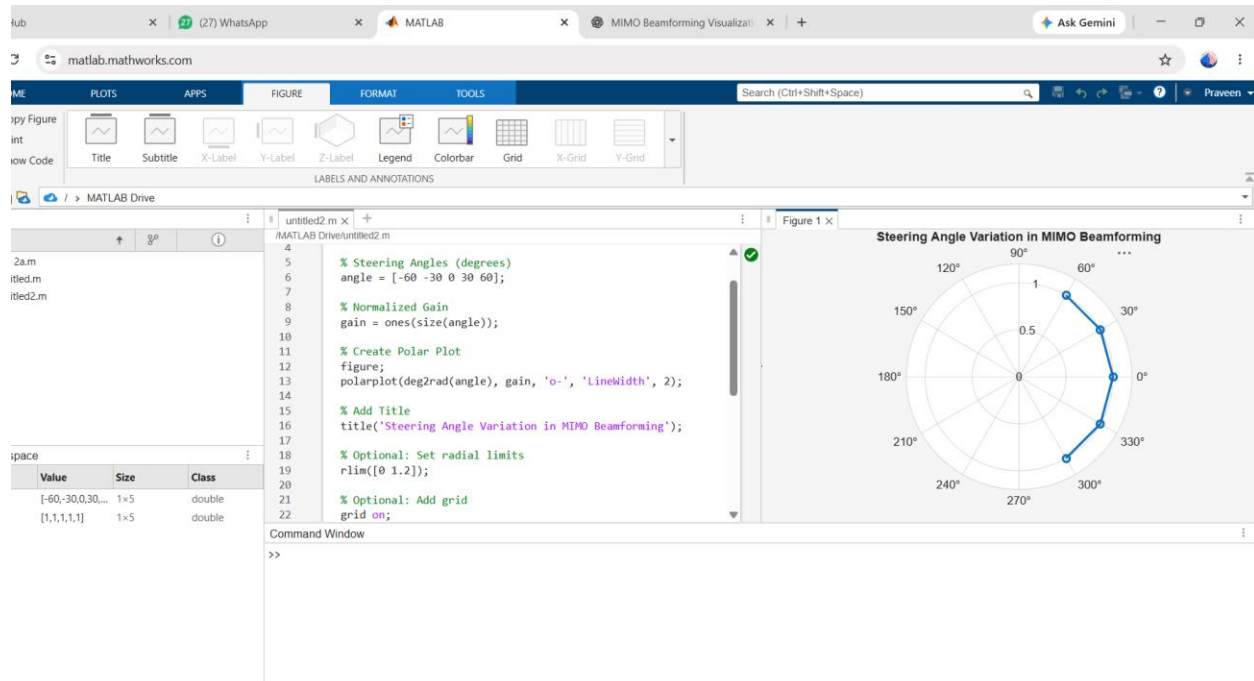
```
title('Steering Angle Variation in MIMO Beamforming');
```

```
% Optional: Set radial limits
```

```
rlim([0 1.2]);
```

```
% Optional: Add grid
```

```
grid on;
```



OBJ 1

clc;

clear;

close all;

% Steering Angles (degrees)

angle = [-60 -30 0 30 60];

% Array Gain (dB)

gain = [9 12 15 12 9];

% Create Bar Graph

figure;

bar(angle, gain);

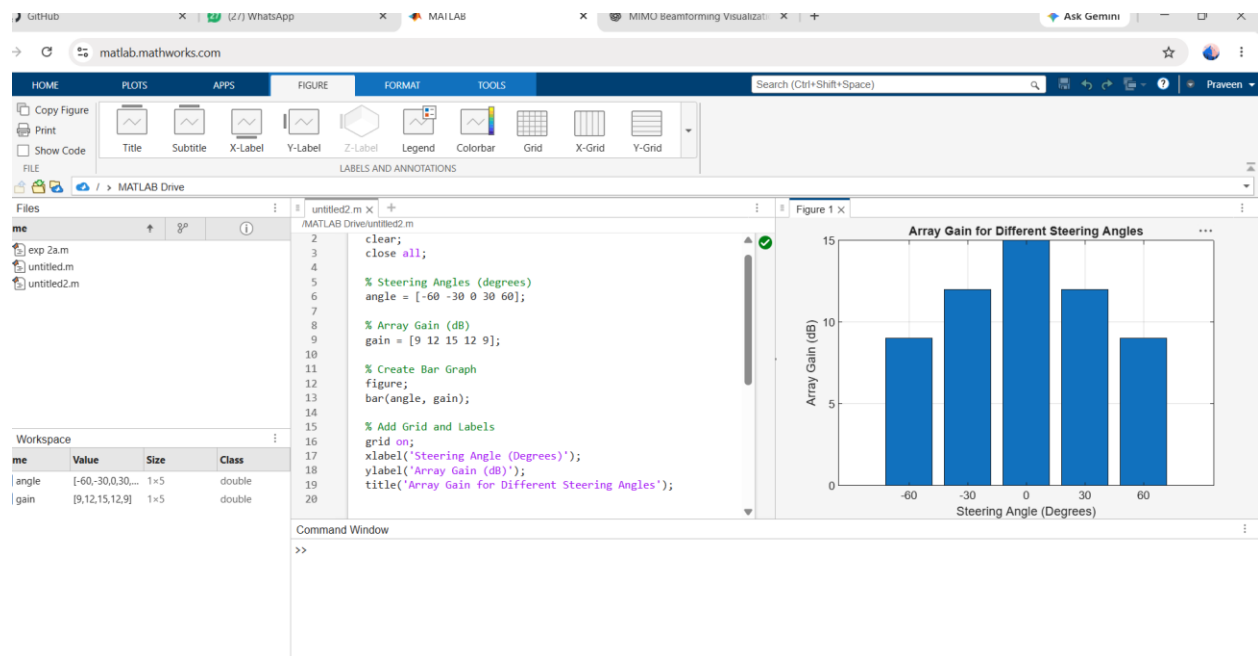
% Add Grid and Labels

grid on;

xlabel('Steering Angle (Degrees)');

ylabel('Array Gain (dB)');

title('Array Gain for Different Steering Angles');



OBJ 2

clc;

clear;

close all;

% Desired Steering Angles (degrees)

desired = [-60 -30 0 30 60];

% Actual Beam Directions (degrees)

actual = [-58 -32 1 29 62];

% Beam Direction Error (Absolute Error)

error = abs(desired - actual);

% Plot Beam Direction Error

figure;

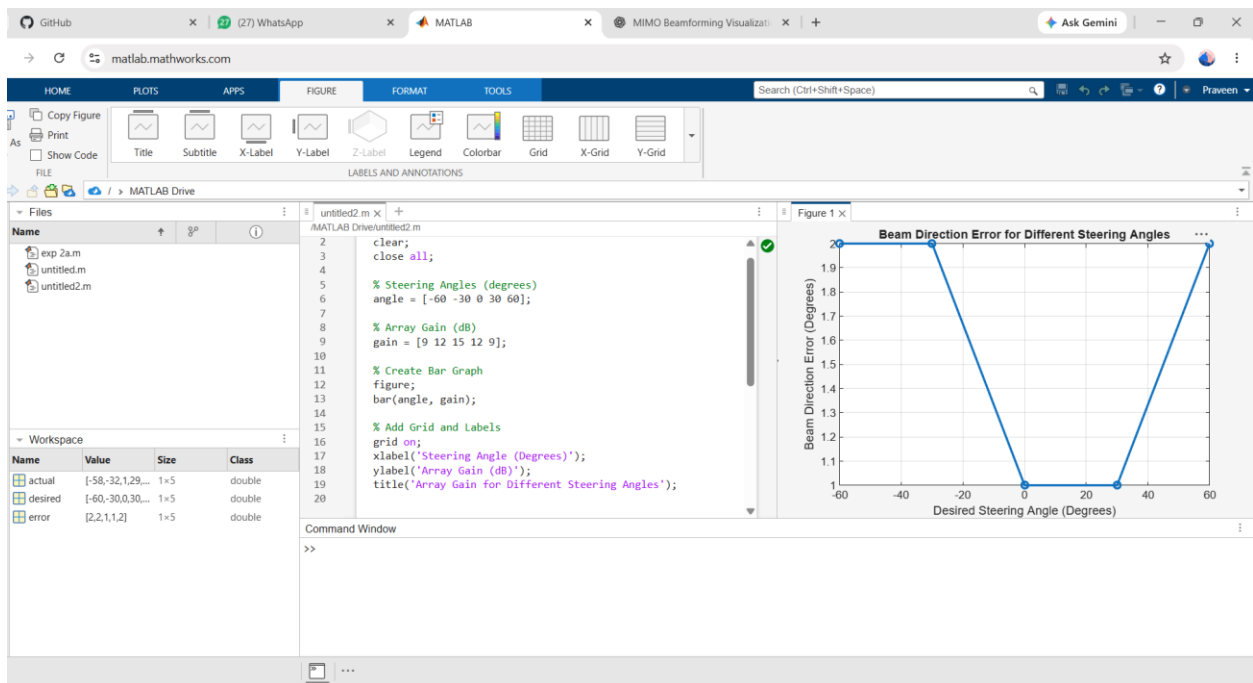
plot(desired, error, 'o-', 'LineWidth', 2);

grid on;

xlabel('Desired Steering Angle (Degrees)');

ylabel('Beam Direction Error (Degrees)');

title('Beam Direction Error for Different Steering Angles');



EXP 9

clc;

clear;

close all;

% SNR values in dB

snr = 0:5:30;

% Convert SNR from dB to linear scale

snr_linear = 10.^(snr/10);

% Calculate Channel Capacity (bps/Hz)

capacity = log2(1 + snr_linear);

% Plot Channel Capacity vs SNR

figure;

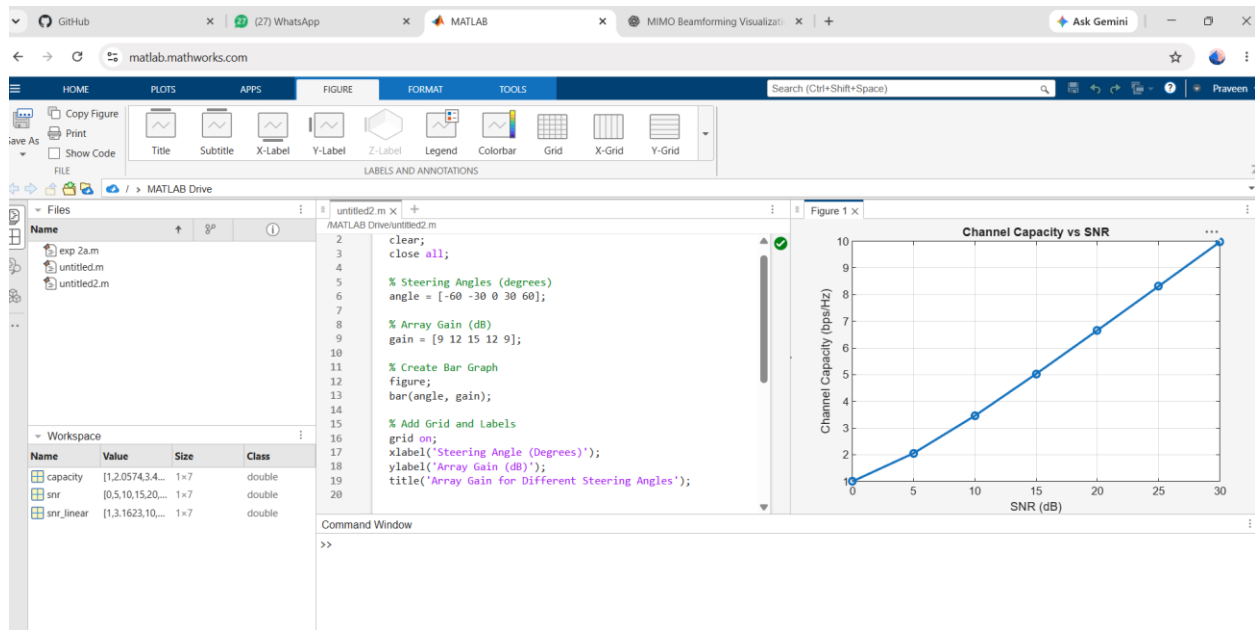
plot(snr, capacity, '-o', 'LineWidth', 2, 'MarkerSize', 6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');



OBJ 2

clc;

clear;

close all;

% SNR values (dB)

snr = 0:5:30;

% Convert SNR from dB to linear scale

snr_linear = 10.^(snr/10);

% Calculate Channel Capacity (bps/Hz)

capacity = log2(1 + snr_linear);

% Plot the graph

```
figure;

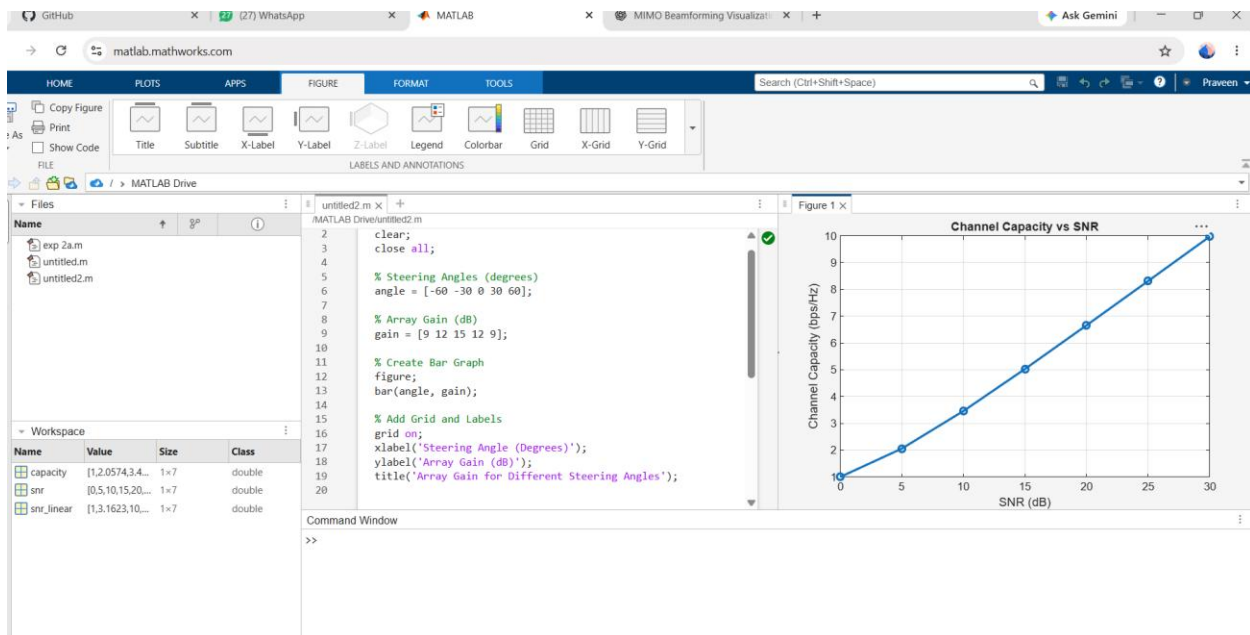
plot(snr, capacity, '-o', 'LineWidth', 2, 'MarkerSize', 6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```



OBJ 3

```
clc;

clear;

close all;
```

```
% Number of Antenna Elements

antennas = [4 8 16 32 64];
```

```
% Antenna Gain (dB)
```

```
gain = 10 * log10(antennas);
```

```
% Plot Bar Graph
```

```
figure;
```

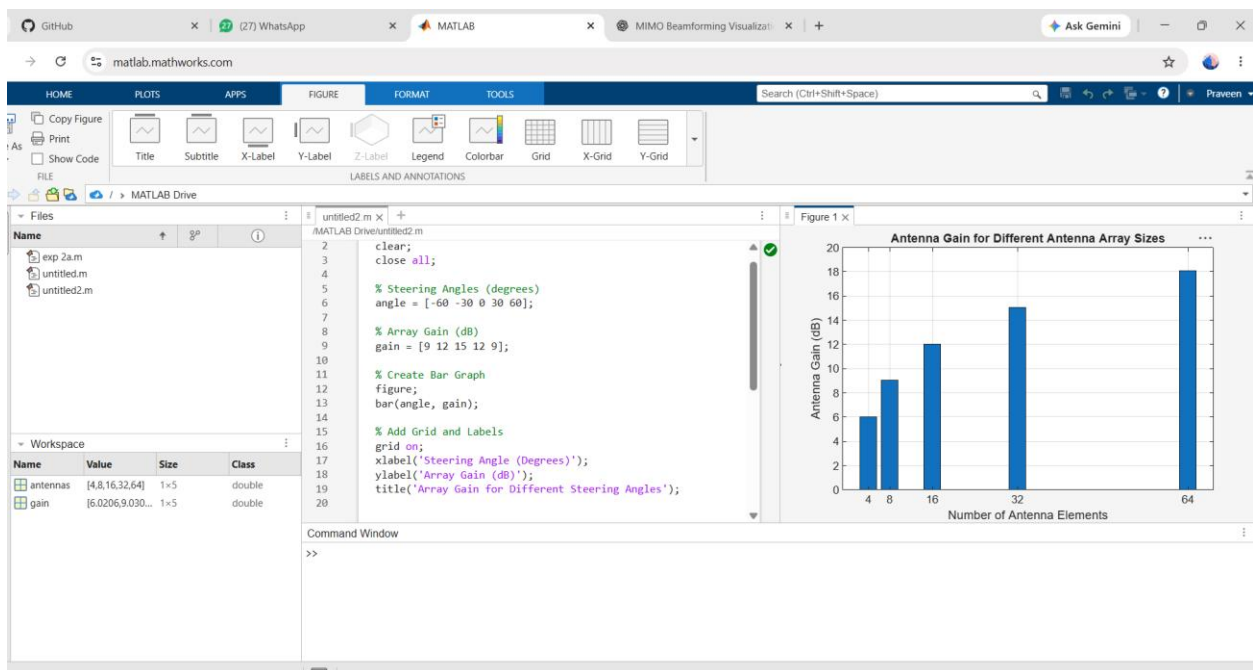
```
bar(antennas, gain);
```

```
grid on;
```

```
xlabel('Number of Antenna Elements');
```

```
ylabel('Antenna Gain (dB)');
```

```
title('Antenna Gain for Different Antenna Array Sizes');
```



EXP 10

```
clc;
```

```
clear;
```

```
close all;
```

```
% Network Slice Types
```

```
slices = {'eMBB','URLLC','mMTC','Private','IoT'};
```

```
% Number of Instances for Each Slice
```

```
count = [1 1 1 1 1];
```

```
% Plot Bar Graph
```

```
figure;
```

```
bar(count);
```

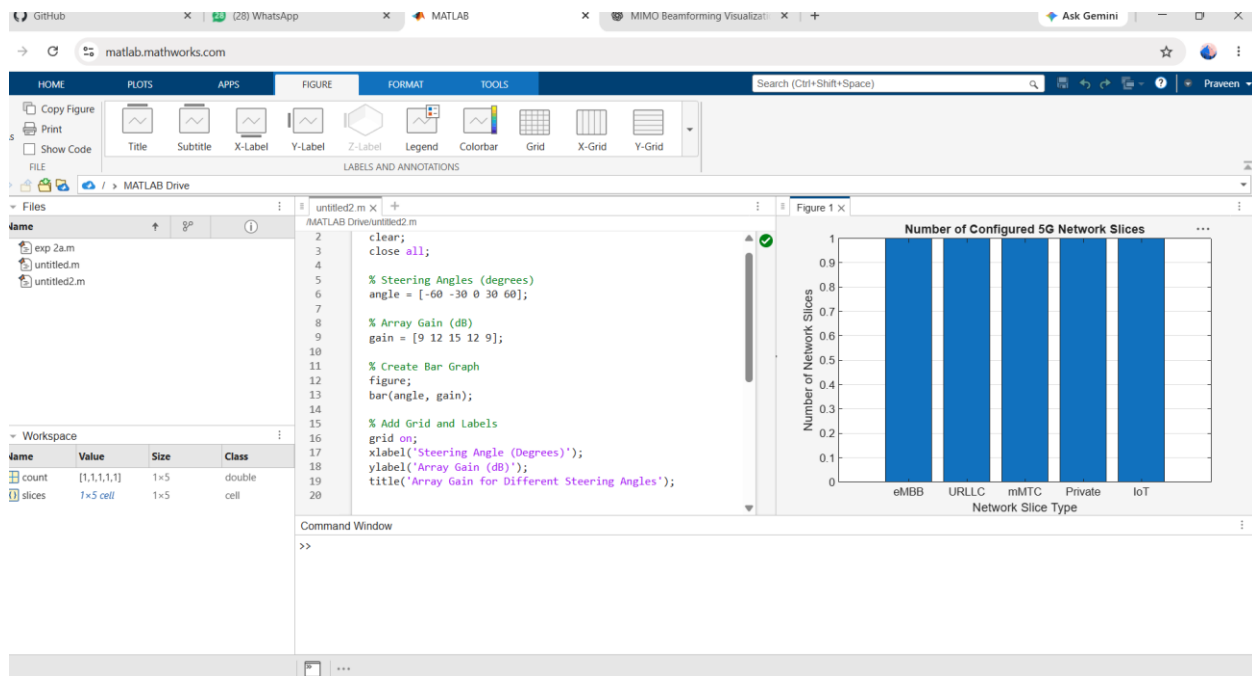
```
grid on;
```

```
set(gca, 'XTickLabel', slices);
```

```
xlabel('Network Slice Type');
```

```
ylabel('Number of Network Slices');
```

```
title('Number of Configured 5G Network Slices');
```



OBJ 2

clc;

clear;

close all;

% Network Slice Types

slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});

% Slice Utilization (%)

utilization = [85 70 60 75 50];

% Plot Bar Graph

figure;

bar(slices, utilization);

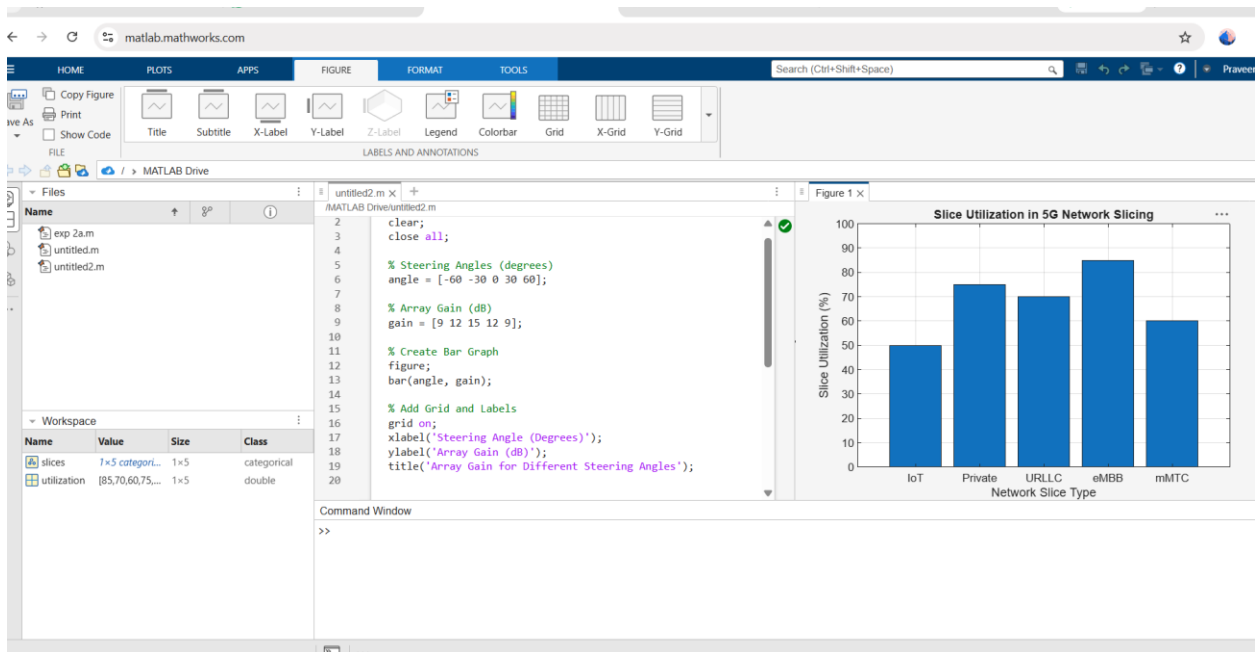
grid on;

xlabel('Network Slice Type');

ylabel('Slice Utilization (%)');

title('Slice Utilization in 5G Network Slicing');

ylim([0 100]);



OBJ 3

clc;

clear;

close all;

% Network Slice Types

slices = [1 2 3 4 5];

% End-to-End Latency (ms)

```
latency = [25 5 40 15 60];
```

```
% Plot Line Graph
```

```
figure;
```

```
plot(slices, latency, '-o', 'LineWidth', 2, 'MarkerSize', 8);
```

```
grid on;
```

```
xticks(slices);
```

```
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
```

```
xlabel('Network Slice Type');
```

```
ylabel('End-to-End Latency (ms)');
```

```
title('5G Network Slicing: End-to-End Latency');
```

```
ylim([0 70]);
```

